Now, Row D contains exactly one zero. Mark that zero by □. Mark other zeros in its column by ×.

Person \ Job	1	2	3	4	5
A	7	3	⊗	5	0
B	[0]	9	4	5	4
C	1	6	6	[0]	4
D	4	3	[0]	⊗	3
E	4	0	2	4	0

Row E contains more than one zero, now proceed column wise. In column 1, there is an assignment. Go to column 2. There is exactly one zero. Mark that zero by □. Mark other zeros in its row by ×.

Person \ Job	1	2	3	4	5
A	7	3	⊗	5	0
B	[0]	9	4	5	4
C	1	6	6	[0]	4
D	4	3	[0]	⊗	3
E	4	[0]	2	4	⊗

There is an assignment in Column 3 and column 4. Go to Column 5. There is exactly one zero. Mark that zero by □. Mark other zeros in its row by ×.

Person \ Job	1	2	3	4	5
A	7	3	⊗	5	[0]
B	[0]	9	4	5	4
C	1	6	6	[0]	4
D	4	3	[0]	⊗	3
E	4	[0]	2	4	⊗

Thus all the five assignments have been made. The Optimal assignment schedule and total cost is

Person	Job	cost
A	5	1
B	1	0
C	4	2
D	3	1
E	2	5
Total cost		9

The optimal assignment (minimum) cost = ₹ 9.

Example 10.9

Solve the following assignment problem.

Task \ Men	1	2	3
P	9	26	15
Q	13	27	6
R	35	20	15
S	18	30	20

Solution:

Since the number of columns is less than the number of rows, given assignment problem is unbalanced one. To balance it, introduce a dummy column with all the entries zero. The revised assignment problem is

Task \ Men	1	2	3	d
P	9	26	15	0
Q	13	27	6	0
R	35	20	15	0
S	18	30	20	0

Here only 3 tasks can be assigned to 3 men.

Step 1: is not necessary, since each row contains zero entry. Go to Step 2.

Step 2 :

Task \ Men	1	2	3	d
P	0	6	9	0
Q	4	7	0	0
R	26	0	9	0
S	9	10	14	0

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